ICS 4111: Embedded Systems & IoT

Semester Project: Deliverable 3

Objective: Transmit and visualise sensor data on cloud platforms

Instructions

1. Create a physical or simulated prototype based on this device architecture: 1 ESP32S connected to 1 MQ-5, 1 DHT22 and 1 LCD.

    FYI: *you can interface this using the physical components at the lab or as a simulation on Wokwi. Physical implementations on this will result in a full score on this marking point.*

2. In your code, include logic for communication to InfluxDB and Grafana as platforms for data storage and visualisation (links for access provided on e-learning). You are free to use any other suitable cloud platforms for the same.
3. On these platforms, ensure you store all sensor data in a time-series database and create a dashboard with at least 3 appropriate visualisations on the collected data.

    Once done, take images of working physical prototypes or save the simulated prototype as a public project on Wokwi and extract the link. For the cloud storage and visualisation platform, generate public links (or screenshots of stored data and the created dashboard, if public link cannot be extracted).

    For submission, generate a Markdown document (.md) in your GitHub repo and include all your content for the deliverable in the document. You can refer to formatting guidelines Markdown docs here. Submit the project repo link to e-learning before the due date. Remember to include evidence of groupwork in your submission.

    Successful submission of this exercise will count towards 2-hour attendance for these classes: ICS 4.1B: 8th July 3.15 – 5.15pm; ICS 4.1A: 9th July 11.15am – 1.15pm; ICS 4.1C: 1.15 – 3.15pm. This will be done by **submitting the deliverable in time and counting all members who appear in the groupwork evidence photo.**